



Jorge Rodas-Silva

Professor and Researcher in Computer Engineering

Associate Professor and researcher at Universidad Estatal de Milagro (UNEMI) and Ph.D. in Computer Engineering from the University of Seville, awarded Cum Laude with International Distinction. His work integrates business intelligence, educational data analytics, digital transformation, and software engineering to advance innovation in higher education. He has more than 15 years of experience in research, teaching, academic innovation, and university management, and currently leads academic process innovation at UNEMI.

ORCID | Scientific profile | jrodass@unemi.edu.ec

38 Publications	99 Citations	5 h-index	66% Open access
---------------------------	------------------------	---------------------	---------------------------

Education

Ph.D. in Computer Engineering 2019 Universidad de Sevilla · Seville, Spain Cum Laude · International Distinction	M.Sc. in Information Technology Management 2011 Universidad Estatal de Milagro · Milagro, Ecuador Honorable Mention · Best Graduate
B.Sc. in Computer Engineering 2010 Universidad Estatal de Milagro · Milagro, Ecuador Honorable Mention · Best Graduate	Business Intelligence and Datawarehouse (BIDWPC) 2013 Escuela Superior Politécnica del Litoral · Guayaquil, Ecuador Professional certification

Research areas

- Business Intelligence and Data Analytics
- Educational Data Mining and Learning Analytics
- Programming Education and Computational Thinking
- Software Engineering and Recommender Systems
- Digital Transformation in Higher Education
- Artificial Intelligence in Education

Academic and professional career

2022 - Present	Director of Academic Process Innovation Vicerrectorado de Formación Académica de Grado · Universidad Estatal de Milagro Academic management
2020 - 2024	Coordinator, M.Sc. in Information Technologies Universidad Estatal de Milagro Graduate program direction
2015 - Present	Associate Professor Universidad Estatal de Milagro Business intelligence and software engineering

2013 - 2014	Research Coordinator Facultad de Ciencias e Ingeniería · Universidad Estatal de Milagro Research
2008 - 2009	Senior Programmer Centro de Especialidades MegaSalud Industry
2007 - 2008	Junior Programmer Gobierno Autónomo Descentralizado Simón Bolívar Industry

Publications

Complete list of 38 publications, ordered from newest to oldest.

2026	A graph-based semi-supervised approach for automatic text corpus emergency events labelling in X (Twitter) Jorge Parraga-Alava, Jorge Rodas-Silva , Jandry Franco, María-José Vélez Big Data Research · 10.1016/j.bdr.2026.100630
	Comparing Academic Performance in Online and Face-to-Face Object-Oriented Programming Courses: An Educational Data Mining Analysis Jorge Rodas-Silva et al. PROCEEDINGS OF 19TH IBERIAN CONFERENCE ON INFORMATION SYSTEMS AND TECHNOLOGIES, CISTI 2024, VOL 3 · 10.1007/978-3-032-12882-9_6
2025	Equity in academic evaluation and quality of university exams: a pattern analysis in higher education J. Cobo, Fabricio Guevara-Viejó, Jorge Rodas-Silva , Cristian Vidal-Silva Formacion Universitaria · 10.4067/S0718-50062025000600135
	Computational Thinking and Academic Performance Across Different Instructional Modalities in Pre-University Courses: A Data-Driven Study Jorge Parraga-Alava, Jorge Rodas-Silva Engineering Reports · 10.1002/eng2.70203
	Bibliometric analysis of teaching strategies for programming courses in higher education, Análisis bibliométrico de estrategias de enseñanza para cursos de programación en educación superior Mariuxi Vinueza-Morales, Jorge Rodas-Silva , Cristian Vidal-Silva, Erika Madariaga-García Formacion Universitaria · 10.4067/s0718-50062025000400039
	Deep Learning for Aphids Detection: A Comparative CNNs Study on Lemon Leaves Images Jorge Rodas-Silva et al. EtcM 2025 9th Ecuador Technical Chapters Meeting · 10.1109/ETCM67548.2025.11304406
	Enhanced Helmet Detection using a CNN and Genetic Algorithm for Personal Protective Equipment Compliance Jorge Rodas-Silva et al. Conference Proceedings IEEE International Conference on Systems Man and Cybernetics · 10.1109/SMC58881.2025.11343106
	Teaching programming in higher education: a bibliometric analysis of trends, technologies, and pedagogical approaches Mariuxi Vinueza-Morales, Jorge Rodas-Silva , Cristian Vidal-Silva, Jorge Córdova-Morán, Edwin Cevallos-Ayón Frontiers in Education · 10.3389/feduc.2025.1525917
	Teaching Programming in Higher Education: Analyzing Trends, Technologies, and Pedagogical Approaches Through a Bibliometric Lens Mariuxi Vinueza-Morales, Jorge Rodas-Silva , Cristian Vidal-Silva International Journal of Advanced Computer Science and Applications · 10.14569/IJACSA.2025.0160106
2024	An effective approach for identifying keywords as high-quality filters to get emergency-implicated Twitter Spanish data Joel Garcia-Arteaga, Jesús Zambrano-Zambrano, Jorge Parraga-Alava, Jorge Rodas-Silva COMPUTER SPEECH AND LANGUAGE · 10.1016/j.CSL.2023.101579
	Applying Educational Programming Language-Based Learning to Enhance the Programming Skills of Non-ICT Engineering College Students Jorge Rodas-Silva , Jorge Parraga-Alava, Mariuxi Vinueza-Morales, Ana Eva Chacón-Luna, J. Cobo Human Behavior and Emerging Technologies · 10.1155/2024/4918351
	Enhancing Electric Power Consumption Prediction Using Machine Learning and Genetic Algorithms Jorge Parraga-Alava, Jorge Rodas-Silva EtcM 2024 8th Ecuador Technical Chapters Meeting · 10.1109/ETCM63562.2024.10746142

2023	A Tuning Approach Using Genetic Algorithms for Emergency Incidents Classification in Social Media in Ecuador Joel García-Arteaga, Jesús Zambrano-Zambrano, Jorge Parraga-Alava, Jorge Rodas-Silva Communications in Computer and Information Science · 10.1007/978-3-031-32213-6_4
	Predicting Academic Performance of Low-Income Students in Public Ecuadorian Online Universities: An Educational Data Mining Approach Jorge Rodas-Silva , Jorge Parraga-Alava Proceedings of the 12th International Conference on Data Science, Technology and Applications · 10.5220/0012086300003541
2022	Comparative Analysis of Methodologies and Technological Tools for Business Intelligence Processes Aimed at Decision-Making María José Guerrero García, Jorge Rodas-Silva Informática y Sistemas · 10.33936/isrtic.v6i1.4522
2021	Use of Web Applications for the Management of Veterinary Clinics and Their Impact on the Improvement of Administrative Processes Anthony Cedenio Ochoa, Axelmann Catuto Murillo, Jorge Rodas-Silva Ecuadorian Science Journal · 10.46480/esj.5.4.174
	Opinion and Sentiment Analysis of Twitter Users during the 2021 Ecuador Presidential Election Jorge Parraga-Alava, Jorge Rodas-Silva , Iván Quimi, Roberth Alcívar-Cevallos PROCEEDINGS OF THE 17TH INTERNATIONAL CONFERENCE ON WEB INFORMATION SYSTEMS AND TECHNOLOGIES (WEBIST) · 10.5220/0010651000003058
2020	Teaching programming with MIT app inventor: A literature review, Enseñanza de programación mediante MIT App Inventor: Una revisión de literatura Mariuxi Vinueza-Morales, Ana Eva Chacón-Luna, Jorge Rodas-Silva , Héctor Serrano Mantilla Proceedings of the LACCEI international Multi-conference for Engineering, Education and Technology · 10.18687/LACCEI2020.1.1.49
2019	RESDEC: Online Management Tool for Implementation Components Selection in Software Product Lines Using Recommender Systems Jorge Rodas-Silva , José Á. Galindo, Jorge García-Gutiérrez, David Benavides 23RD INTERNATIONAL SYSTEMS AND SOFTWARE PRODUCT LINE CONFERENCE(SPLC 2019), VOL B · 10.1145/3307630.3342390
	Automated analysis of two-layered feature models with feature attributes Michael Lettner, Jorge Rodas-Silva , José Á. Galindo, David Benavides Journal of Computer Languages · 10.1016/j.col.2019.01.005
	Selection of Software Product Line Implementation Components Using Recommender Systems: An Application to Wordpress Jorge Rodas-Silva , José Á. Galindo, Jorge García-Gutiérrez, David Benavides IEEE ACCESS · 10.1109/ACCESS.2019.2918469
	Using Alice software as a tool for programming learning: A literature review, El uso del software Alice como herramienta para el aprendizaje de programación: Una revisión de literatura Jorge Antonio Córdova Morán, Jorge Rodas-Silva , Mariuxi G. Vinueza Morales Proceedings of the LACCEI international Multi-conference for Engineering, Education and Technology · 10.18687/LACCEI2019.1.1.161
2018	Selection of Deployment Configurations Using Recommender Systems on Android Jorge Rodas-Silva , José A. Galindo, David Benavides, R. Soriano RISTI - Revista Ibérica de Sistemas e Tecnologias de Informação
	RESDEC: A Prototype Tool for the Selection of Deployment Configurations Based on Recommender Systems Jorge Rodas-Silva et al. Actas de las 23rd Jornadas de Ingeniería del Software y Bases de Datos, JISBD 2018
2017	Programming with the SCRATCH + CARAMBA tool. A significant learning experience, Programación con la herramienta SCRATCH +CARAMBA. Una experiencia de aprendizaje significativo Jorge Rodas-Silva et al. Proceedings of the LACCEI international Multi-conference for Engineering, Education and Technology · 10.18687/LACCEI2017.1.1.268
2016	Using Attributed Feature Models for Testing Highly Configurable Systems: First Steps Mariuxi Vinueza, Jorge Rodas-Silva , José Ángel Galindo Duarte, David Benavides Actas de las 21st Jornadas de Ingeniería y Bases de Datos, JISBD 2016
	The use of feature models with attributes for testing variability intensive systems: First steps Jorge Rodas-Silva et al. Actas De Las 21st Jornadas De Ingeniería y Bases de Datos, JISBD 2016
	Towards the Use of Recommender Systems in Highly Configurable Systems Jorge Rodas-Silva , Javier Olivares, José Ángel Galindo Duarte, David Benavides Actas de las 21st Jornadas de Ingeniería y Bases de Datos, JISBD 2016
	Towards the use of recommender systems in variability intensive systems Jorge Rodas-Silva et al. Actas De Las 21st Jornadas De Ingeniería y Bases de Datos, JISBD 2016

Body Bias usage in UTBB FDSOI designs: A parametric exploration approach
Jorge Rodas-Silva et al.
SOLID-STATE ELECTRONICS · 10.1016/j.sse.2015.11.019

2015

Business Intelligence System and Its Impact on Decision-Making in Telecommunications Companies in La Troncal, Ecuador
Jorge Luis Rodas-Silva, Manuel Guillermo Rodríguez-López, Jesennia Cárdenas-Cobo
Copérnico

Towards testing variability intensive systems using user reviews
Jorge Rodas-Silva, David Méndez-Acuña, José Á. Galindo, David Benavides, Jesennia Cardenas
2015 10TH COMPUTING COLOMBIAN CONFERENCE (10CCC) · 10.1109/ColumbianCC.2015.7333410

Strategic Planning Through ICT in Rural Decentralized Autonomous Governments of Milagro Canton
Mariuxi Geovanna Vinueza Morales, **Jorge Rodas-Silva**, Ana Chacón Luna
Ciencia UNEMI · 10.29076/issn.2528-7737vol8iss15.2015pp40-49p

Standards That Contribute to the Development and Delivery of High-Quality Software Products
Ana Chacón Luna, **Jorge Luis Rodas-Silva**, Mariuxi Vinueza Morales
Ciencia UNEMI · 10.29076/issn.2528-7737vol8iss13.2015pp90-99p

Evaluation and Mitigation of Aging Effects on a Digital On-Chip Voltage and Temperature Sensor
Jorge Rodas-Silva et al.
PROCEEDINGS 2015 25TH INTERNATIONAL WORKSHOP ON POWER AND TIMING MODELING, OPTIMIZATION AND SIMULATION

2014

Digital Management Systems to Improve Academic Processes in Educational Institutions
Jorge Rodas-Silva, Jesennia Cárdenas-Cobo
Universidad, Ciencia y Tecnología

E-Commerce: A Perspective from SMEs for Developing Strategies to Promote Economic and Business Growth in the City of Milagro
Jorge Luis Rodas-Silva, Ana Eva Chacón Luna, Mariuxi Geovanna Vinueza Morales
ECA Sinergia

2012

Study to Determine the Use and Application of ICT in Teaching and Learning Processes by Educators in Milagro and Neighboring Cantons
Jorge Luis Rodas-Silva
Ciencia UNEMI · 10.29076/issn.2528-7737vol5iss8.2012pp79-92p

Awards

2019	Ph.D. Cum Laude with International Distinction	Unanimous decision · Universidad de Sevilla
2011	Best M.Sc. Graduate in IT Management	Honorable Mention · First cohort · Universidad Estatal de Milagro
2006	Best Graduate in Computer Systems Engineering	Honorable Mention · Universidad Estatal de Milagro

Metrics updated: 2026-08-28. Sources: ORCID, Scopus, OpenAlex, Latindex, Crossref.